

SIMATS ENGINEERING

ASSESSMENT TOOL 3: class test

Course name: Computer vision with opencv

Course code: DSA02

CO2: Apply image enhancement and segmentation techniques for extracting meaningful from images.

Assessment Tool Weightage: 10%.

NAME: SYED THOUQEER AHMED. A

Reg No: 190304096.

1) An image appears too bright. Explain and justify the most suitable gray level transformation to improve visibility.

* Method: Power-law (Gamma) Transformation ($\gamma > 1$).

* Explanation: Uses the formula $s = c \cdot r^\gamma$ to map high-intensity pixels to a darker range. This compresses overly bright regions and darkens the midtones, restoring washed-out details.

2) An image has poor contrast. Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

Method: Histogram Equalization
~~~~~

Explanation: Computes the probability distribution of pixel intensities and flattens the histogram. This spreads pixel values across the entire dynamic range, intensity maximizing overall contrast.

- 5) An image shows uneven brightness across regions. Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

Method: Homomorphic Filtering  
~~~~~

Explanation: A frequency-domain technique that separates illumination from reflection. It suppresses low-frequency components (uneven lighting) while amplifying high frequency components.

- 6) A noisy image contains random intensity variations. Explain and Justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

* Method: Gaussian Smoothing Filter
~~~~~

\* Explanation: Replaces each pixel with a weighted  
~~~~~

local Average based on Gaussian (bell) curve. This mathematically cancels out zero-mean random noise while creating a smoother, more natural-blur than a standard mean filter.

- 7) An image requires noise reduction without significant loss of details. Describe and apply an appropriate filtering approach, explanation how it balances noise removal and detail preservation.

* Method: Bilateral Filter

* Explanation: A non-linear filter that weights nearby pixels based on both spatial closeness and intensity similarity. This prevents the filter from averaging across sharp boundaries, thereby smoothing flat regions while preserving edges.

- 8) An image appears blurred after smoothing. Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

* Method: Unsharp masking

* Explanation: Subtracts a blurred version of the image from the original to isolate the high-frequency

edges (creating a mask). Adding a scaled version of this mask back to the original image amplifies the edges and restores crispness.

- 15) An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.

* Method: Global Thresholding (Otsu's Method)
~~~~~

\* Explanation: Automatically finds the optimal threshold value by minimizing the variance within the foreground and background pixel classes, cleanly separating the bimodal image into black and white.

- 16) Detected edges in an image appear broken. Explain and justify a method to improve edge continuity, including its role in post-processing.

\* Method: morphological closing (dilation followed by erosion)  
~~~~~

* Explanation: Dilation first expands the white edge pixels to bridge and fill small gaps. The immediate erosion then shrinks the boundary

back to its original thickness, resulting in a continuous connected edge.

21) An image suffers from excessive brightness variation between regions. Explain and Justify a suitable regions histogram processing technique to normalize intensity distribution, describing its working and Expected outcome.

* Method: contrast limited adaptive Histogram equalization (CLAHE)

* Explanation: Divides the image into small grid tiles and equalizes the histogram locally rather than globally. It clips the histogram to prevent noise amplification, perfectly balancing extreme shadows and highlights.

22) An image contains salt and pepper noise. Explaination and Justify the most appropriate spatial mechanism and effectiveness in noise removal.

* Method: Median Filter.

* Explanation: Sorts the pixel values in a local

neighborhood and replaces the center pixel with the median value - Salt (255) and pepper (0) are extreme outliers, they are completely ignored removing the noise without blurring edges.
